

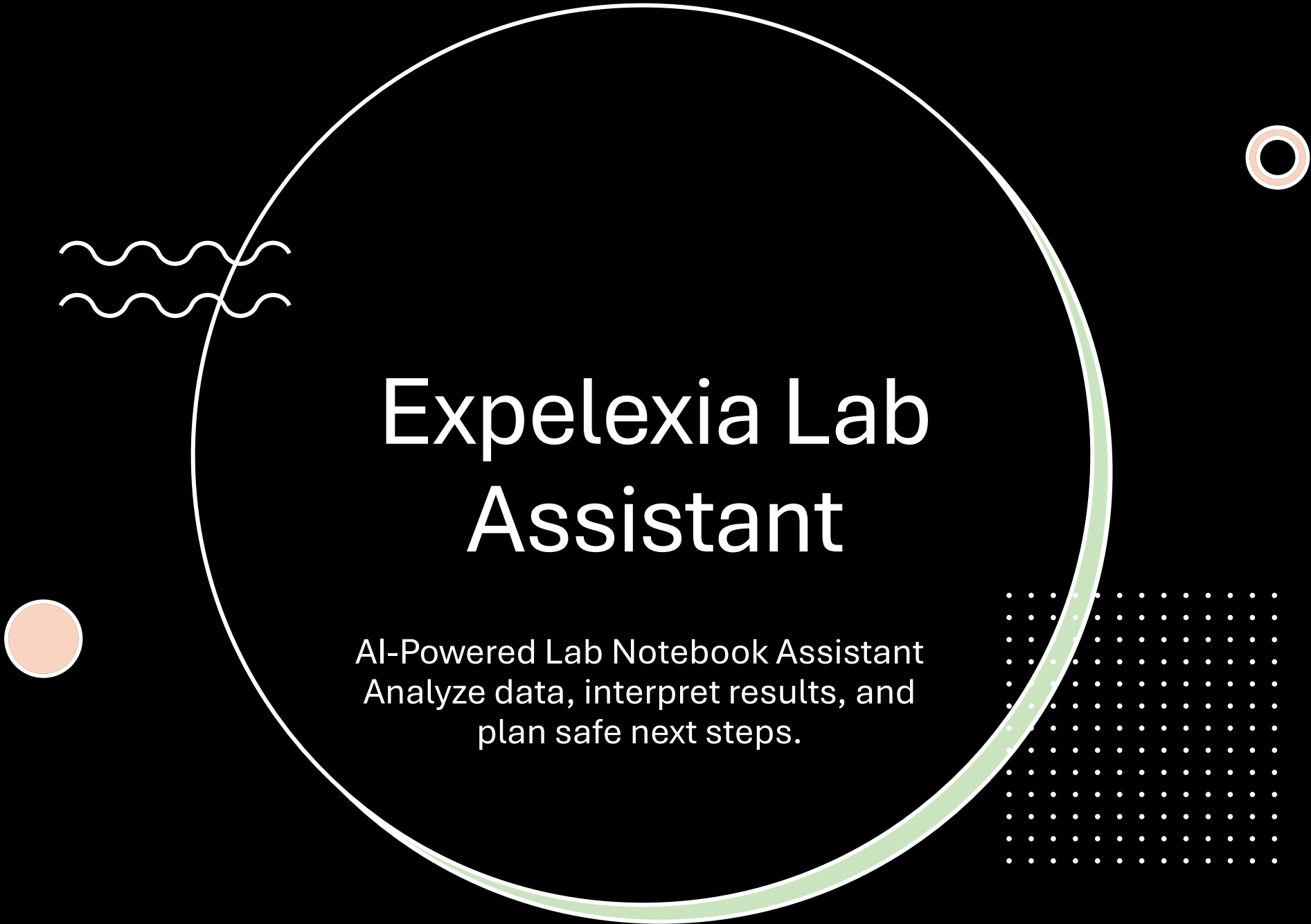


Microsoft AI Innovation Challenge 2026

From raw data to intelligent
decisions Design



Microsoft



Expelexia Lab Assistant

AI-Powered Lab Notebook Assistant
Analyze data, interpret results, and
plan safe next steps.

NeuraForge Team



Henok Tariku

- **Team Member and Leader**
- **Role:** Junior Data Analyst & Software Engineer
- **Skills:** Data Analysis, Software Engineering, API Integration, Machine Learning, Automation, Python, Power BI, Cloud (Azure)
- **Location:** Moscow, Russia

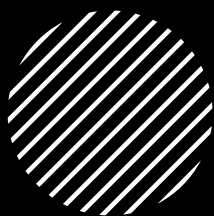


Rocio Herrera

- **Team Member**
- **Role:** Cybersecurity & AI Enthusiast
- **Skills:** Cybersecurity, Artificial Intelligence, Problem Solving, Data Analysis, Risk Assessment
- **Location:** Los Angeles, USA

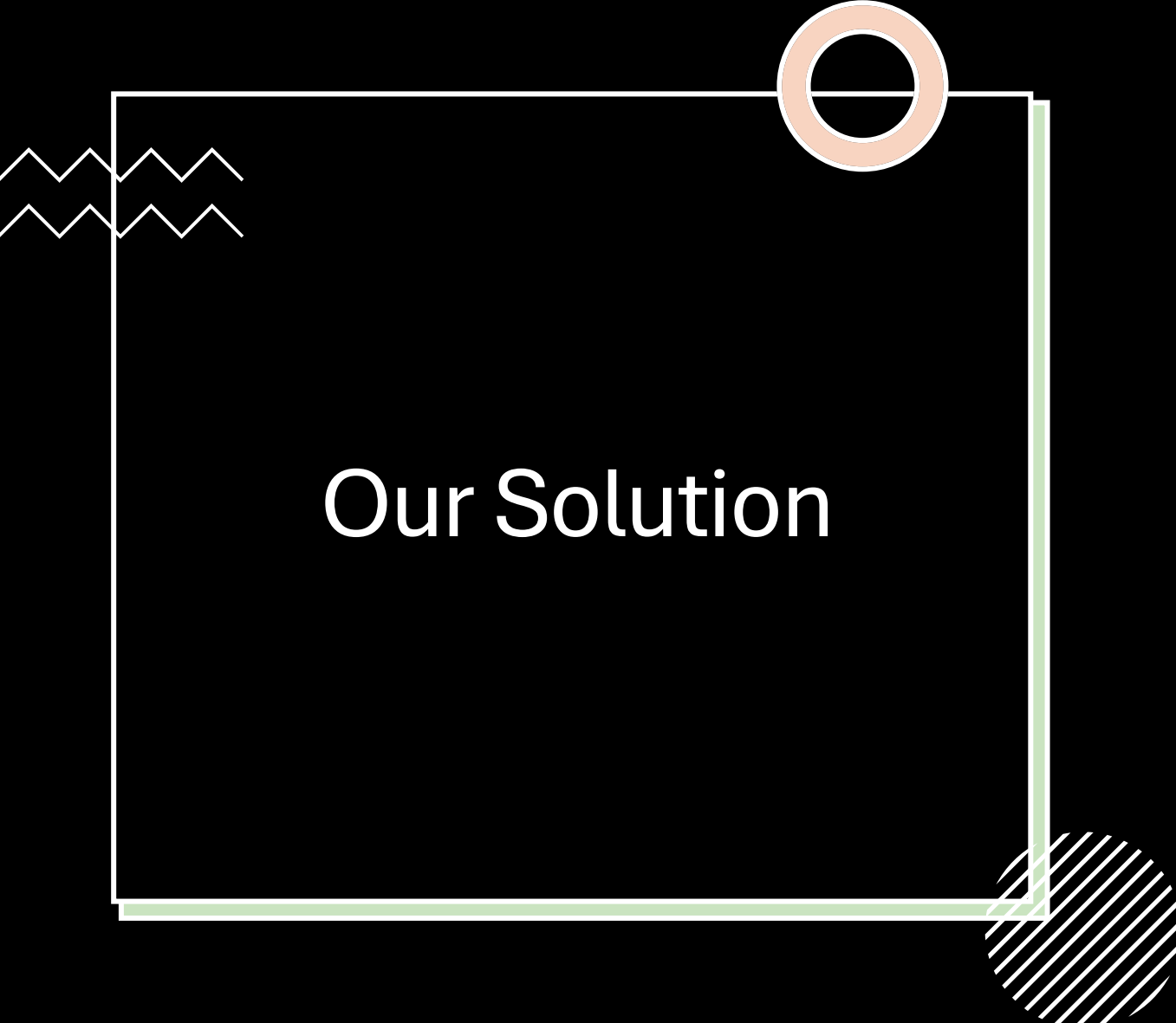


The Problem



- IoT systems generate massive datasets
- Manual analysis is slow and error-prone
- No easy way to generate insights & reports

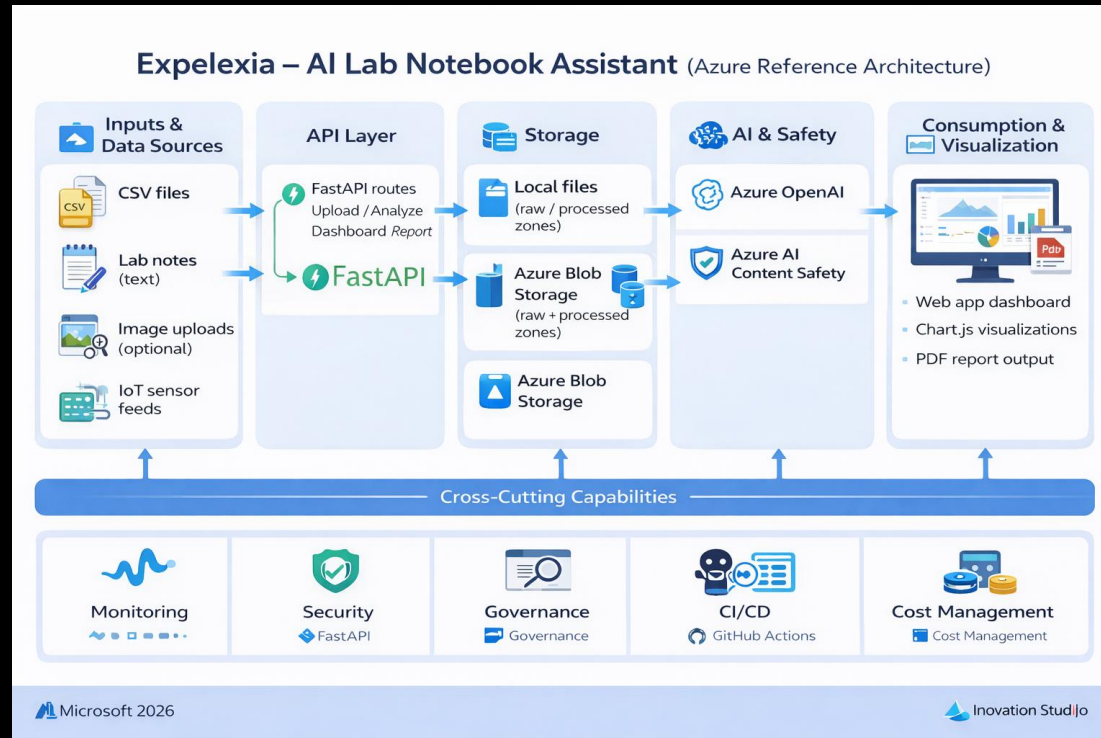




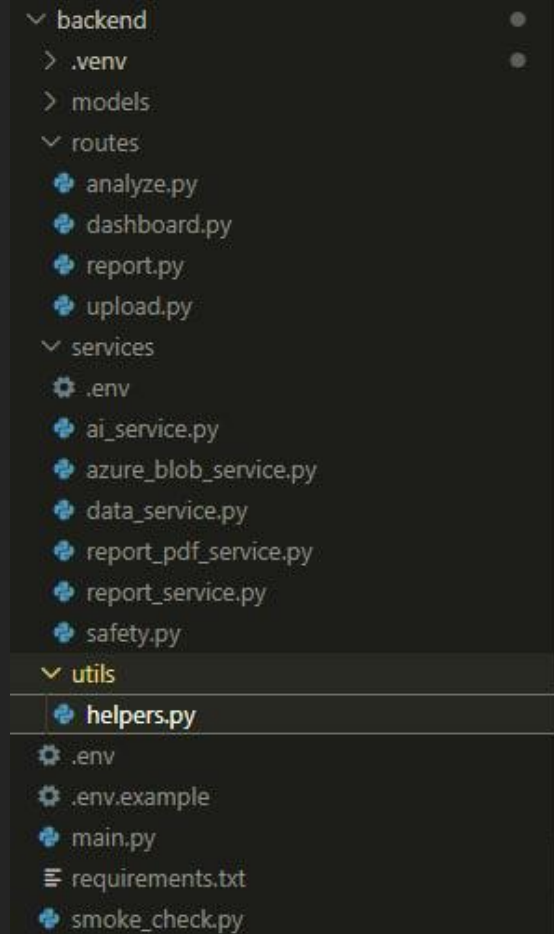
Our Solution

- AI-powered data analysis platform
- Automated processing → insights → reports
- Simple interface for complex data

Key Features



- Upload CSV/PDF
- AI analysis
- Dashboard
- Report generation



Backend Overview

- FastAPI backend
- API routes:
 - ❖ /api/upload
 - ❖ /api/analyze
 - ❖ /api/report
- Azure + AI services

Backend Code Example

```
ckend > main.py > ...  
You, 10 hours ago | 1 author (You)  
1 from fastapi import FastAPI  
2 from fastapi.middleware.cors import CORSMiddleware  
3 from routes.upload import router as upload_router  
4 from routes.analyze import router as analyze_router  
5 from routes.report import router as report_router  
6 from routes.dashboard import router as dashboard_router  
7 app = FastAPI(title="Expelexia Lab Assistant")  
8  
9 app.add_middleware(  
10     CORSMiddleware,  
11     allow_origins=[  
12         "*",  
13     ],  
14     allow_credentials=True,  
15     allow_methods=["*"],  
16     allow_headers=["*"],  
17 )  
18  
19 # Test route  
20 @app.get("/")  
21 def root():  
22     return {"message": "Backend is running"}  
23  
24 # Register upload route  
25 app.include_router(upload_router, prefix="/api")  
26 app.include_router(analyze_router, prefix="/api")  
27 app.include_router(report_router, prefix="/api")  
28 app.include_router(dashboard_router, prefix="/api")
```



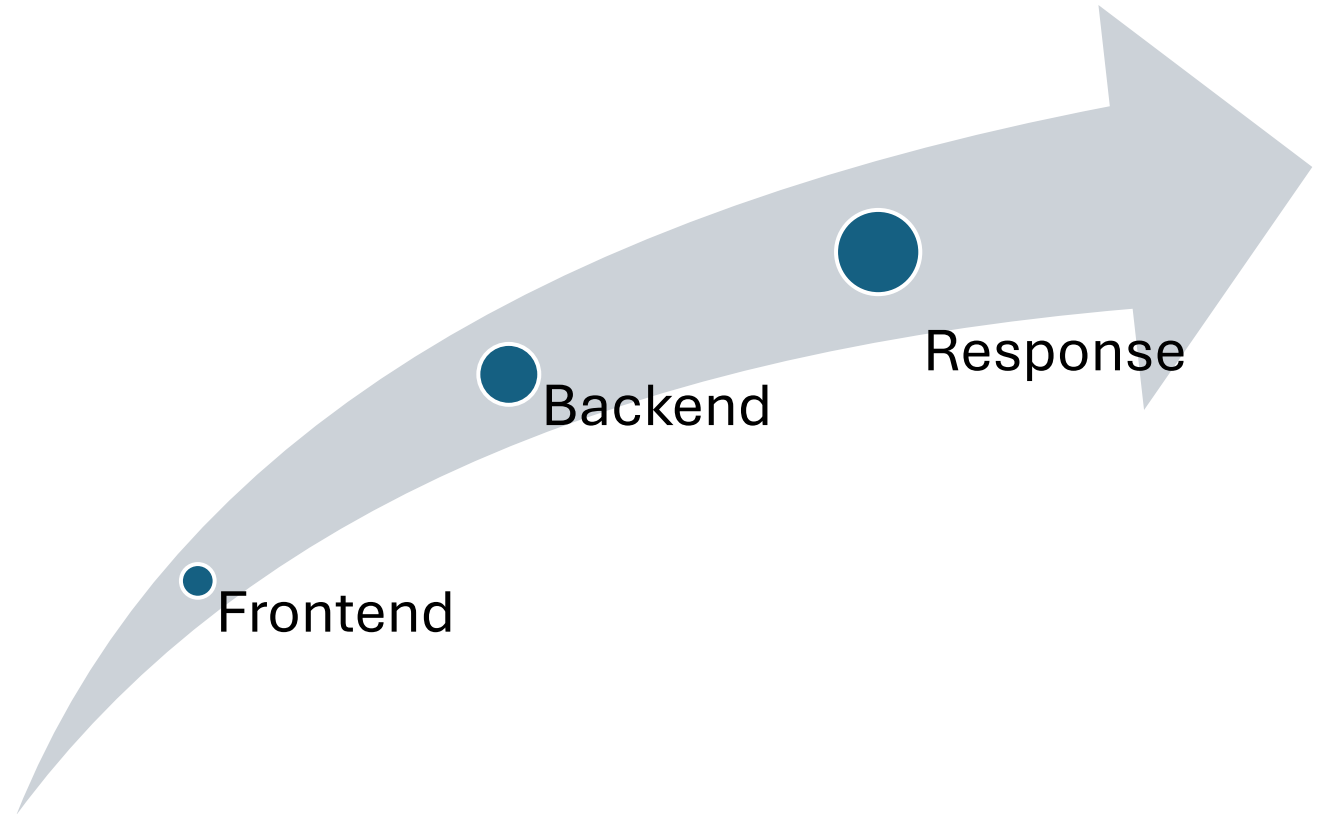
Frontend Overview

- Next.js + React
- UI for upload & dashboard

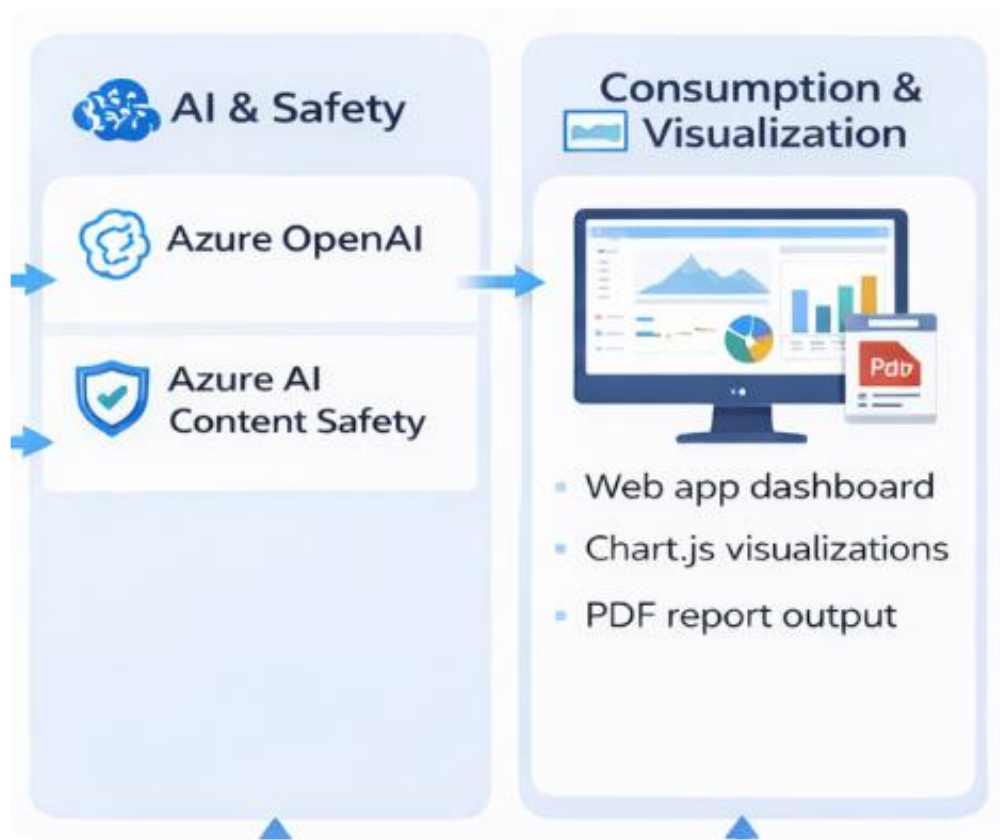
```
frontend > utils > TS api.ts > ...
You, 7 hours ago | 1 author (You)
// Use Render backend only
1  const API_BASE = 'https://backend-update-1-uloa.onrender.com/api';
2  let activeApiBase = API_BASE;
3
4
5  const FILE_HISTORY_KEY = 'expelexia-file-history';
6  const LAST_FILE_KEY = 'expelexia-last-file';
7  const NEW_FILE_KEY = 'expelexia-new-file';
8  const ANALYSIS_HISTORY_KEY = 'expelexia-analysis-history';
9
10 type UploadResponse = {
11   filename: string;
12   azure_blob_url: string;
13 };
14
15 type AnalyzeResponse = {
16   summary_json?: {
17     metadata?: {
18       rows?: number;
19       columns?: number;
20     };
21     stats?: Record<string, { mean?: number }>;
22   };
23 };
24
25 type DashboardResponse = {
26   data_type?: string;
27   text_explanation?: string;
28   recommendations?: string;
29   data_change_explanation?: string;
30   chart_explanation?: string;
31   technical_review?: string;
32   technical_recommendations?: string;
33   document_notes?: string;
34   microsoft_link?: string;
35   table_preview?: Array<Array<string | number>>;
36   charts_data?: {
37     line?: Array<{ metric: string; points?: Array<{ x: string | number; y: number }> }>;
38     pie?: Array<{ label: string; value: number }>;
```

API Communication

- Frontend calls backend:
`fetch(${BASE_URL}/api/upload)`
- Backend processes & returns JSON



AI Analysis



AZURE OPENAI
ANALYZES DATA



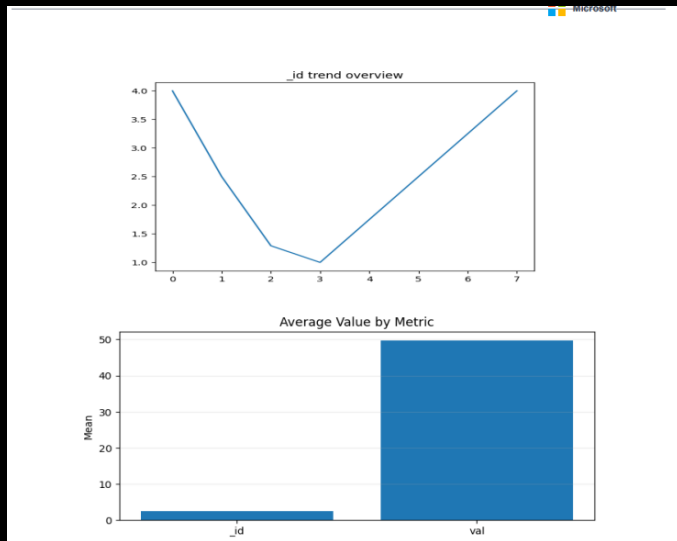
DETECTS PATTERNS &
ANOMALIES



GENERATES
RECOMMENDATIONS

Dashboard

- Charts & metrics
- Data visualization
- Real-time insights



Expelexia Lab - Data Analysis Report

File Name: e123e70c-9d13-461e-8715-f06ec41dd3cf.csv

Date of Analysis: 2026-03-26 15:46

Prepared For: Expelexia (AI Analytics Program)

Project: Expelexia Lab

This report provides a comprehensive analysis and AI-generated insights based on the uploaded data.

Executive Summary

This report summarizes analysis results for e123e70c-9d13-461e-8715-f06ec41dd3cf.csv using the same data shown in the website dashboard.

The dataset contains 4 records across 4 variables.

Main areas needing attention: _id: high variance, val: high variance.

Observed trend signals: _id up, val up.

Recommendations focus on practical monitoring, early detection of unusual changes, and improving reliability of future data collection.

Data Overview

Measure	Value	What it means
Number of records	4	How many entries were analyzed.
Number of variables	4	How many fields were evaluated.
Missing values	{'_id': 0, 'val': 0}	Missing entries can reduce re

Report Generation

- PDF reports generated automatically
- Uses:
 - Reportlab
 - pandas

Deployment

- Backend:

- Render

- Gunicorn + Uvicorn

- Frontend:

- Netlify

- URLs: `https://backend-update-1-uloa.onrender.com/api`

- Backend: onrender

- Frontend: netlify



Challenges & Fixes

- 
- API not connecting → fixed URL
 - CORS issues → middleware
 - Azure auth errors → env variables
 - Deployment issues → gunicorn fix
- 

Final / Conclusion

- Expelexia automates lab data analysis
- AI-powered insights
- Full-stack deployed system
- GitHub link:
<https://github.com/Adam-7th/Expelexia>

